IT Risk Management

ES234423 – 3 Credits
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Week 7: IT Trends and Issues in AI Era

Eko Wahyu Tyas Darmaningrat

tyas.darmaningrat@its.ac.id (082331106699)

Mark Your Attendance!

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Learning Objectives

Comprehend information security trends and their potential threats.

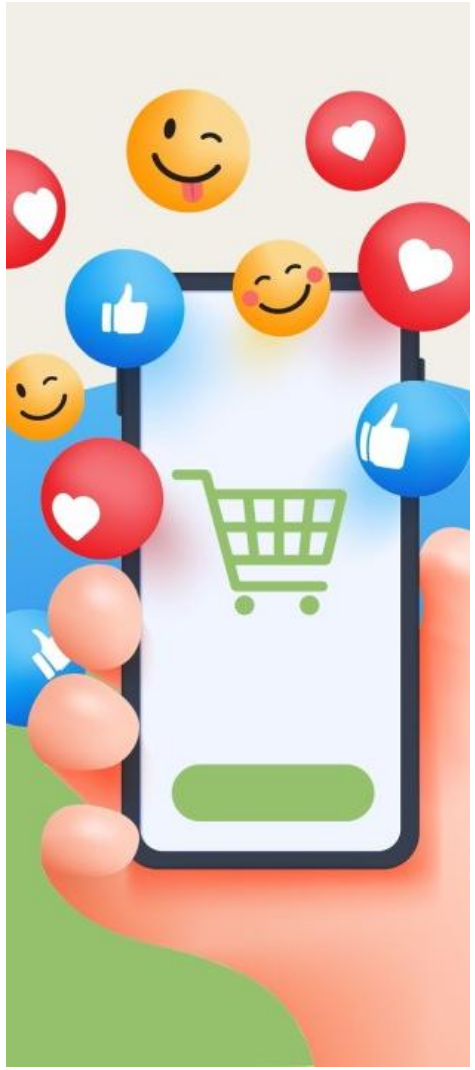
Outline

- Indonesia as a Digital Giant
- The New Digital Explosion (What's Different Now)
- Organisations Are Transforming Too
- Information Security Issues in AI Era
- Class Activity

Indonesia as a Digital Giant

- Indonesia is now one of the largest digital markets in the world.
- Over **212 million internet users** (APJII, 2024).
- Almost everyone:
  - uses smartphones
  - is active on social media
  - is constantly connected
- This means: We are not just using technology anymore. We are **living inside a digital ecosystem**.

The New Digital Explosion (What's Different Now)



- In the past, we talked about e-commerce. Today, it has evolved into something much bigger.
- **Social Commerce:** People now shop directly from social media platforms.
  - TikTok + shopping = one experience (TikTok Live → instant checkout)
  - Instagram + influencers = direct selling (Influencers act as digital sales channels).
- Insight: Entertainment and commerce are now fully integrated.

The New Digital Explosion (What's Different Now) - (2)

- **Creator Economy:** Anyone can now become a business.
  - Content = income
  - Personal brand = career asset
- Students today can:
  - become content creators
  - build audiences
  - monetize their skills
- This changes the job market: Your CV is no longer your only professional identity.

The New Digital Explosion (What's Different Now) - (2)

- **AI Adoption (The Biggest Shift):** Today, AI tools are used daily.
  - ChatGPT → learning, writing, ideation
  - Microsoft Copilot → productivity & coding
  - Google Gemini → analysis & research

- AI is not the future. AI is already part of our daily workflow.

Organisations Are Transforming Too

- **Banking Sector**
  - Banks use AI for fraud detection or anomaly detection.
  - AI can identify suspicious patterns faster than humans.
- **E-Commerce Platforms**
  - AI is used for product recommendations and user personalisation
  - Ever wondered why ads feel “too accurate”? Because AI learns from your behavior.
- **Government**
  - Governments are adopting AI chatbots for public services and data-driven decision systems.
  - Goal: faster services and more efficient operations.

The Big Insight

We need to realise something critical:

- We are not just users of technology. We are part of the system.
  - Our data → fuels AI.
  - Our behavior → trains algorithms.
  - Our decisions → are influenced by platforms.
- We are moving from a Digital Society to an Intelligent Society, where humans no longer just use technology but coexist with AI.

Information Security Issues in AI Era

In the past, information security problems were quite simple: viruses, hacking, password leaks. But today?

- Deepfake (video & voice)
- AI-generated misinformation (fake news, biased narratives)
- Data privacy leakage (prompt leakage! I.e. leaking sensitive information through our prompts.)
- Shadow AI (Employees use AI without company approval/security control)
- Social engineering 2.0 (AI-powered scam)
- Algorithmic manipulation (TikTok & IG use algorithms to decide what we see and influence our opinions)

Class Activity 1: AI Security Risk Analysis Report

- Analyse **one real-world case** related to security risks in the AI & social media era.
- Deliverables: Report (8-10 pages), don't forget to use the CA template.
- **Required Structure**
  1. **Case Description:** What happened? Who was involved? Provide link to the original case.
  2. **Technology Analysis:** What technology was used? How did AI play a role?
  3. **Risk Identification:** Identify at least **3 risks**
  4. **Impact Analysis:** Impact on individuals, organisations, or society.
  5. **Mitigation Strategy:** How can this be prevented? Include technical, policy, or awareness solutions.
  6. **Reflection (Most Important):** What did you learn? Have you experienced something similar?

Class Activity 2: Digital Survival Guide in AI Era

- Based on your knowledge and experiences, create 5–7 **practical rules** for staying safe in the AI era.

- Submit your work on myITS Classroom. Submission format: Week 7_NRP.pdf

Have a wonderful week!